CONFIDENTIAL · INVESTOR BRIEF

The Verification Gap

Generation is solved. Verification is not — and it is the bottleneck every leading AI coding tool leaves open. This brief lays out the evidence, compares the field, and states plainly what is a real edge and what is not yet proven.

Valiant Verification · morpheus.nz/verification · October 2026

All claims cited to primary vendor documentation or the live system. Financial figures are third-party and flagged.

Executive summary

The scarce resource in AI-built software has moved downstream — from *generating* code to *trusting* what was generated.

Anthropic, running an AI-native engineering organisation, said it directly: **"Verification, code review, and security became the bottlenecks once generation was cheap."**

We surveyed the twelve leading systems. **Ten of twelve leave verification optional.** "Done" means "the model stopped", not "verified". The mechanism they all share is a **hook** — a shell command the model may or may not run. A hook the model can skip is not a gate.

Morpheus ships the opposite: verification as a **structural gate** — required checks matched by name, a reviewer that can never grade its own work, and every guard **proven to fail** on a seeded fault. That discipline is the one thing the funded field does not ship.

The honest read. This is real as a *discipline* and as a *running system* — and it is **not yet** real as *traction*. Pre-revenue, solo operator, 7 accounts. An investment here buys the methodology, the first-mover position on the trust layer, and the differentiation — **not** a revenue curve. We state this on every page, because an overstated number is the one failure that cannot be walked back.

The market moment

Every serious tool runs the same loop — **plan → edit → run → verify** — a harness around a model looping over tools. The model is rented and identical across vendors. The difference between them is whether the **verify** step is something the system *enforces*, or something the model *may skip*.

> **Nobody owns "trust".** The incumbents optimise for demo velocity. Verification, ownership, and model-freedom are the three things they under-serve. Standards are open (AGENTS.md + MCP were donated to the Linux Foundation); the *discipline* that makes them trustworthy is the scarce, unclaimed asset.
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> Market context — third-party figures, treated as point-in-time and not relied on: Cursor's parent Anysphere ~$60B (reported acquisition), Devin ~$25B (Sacra), Factory ~$5B post-money on ~$420M raised (Sacra). The point is not the valuations — it is that *none of them own the trust layer*.

The receipts — twelve tools, ten leave verification optional

Compiled from primary vendor documentation and engineering posts (2025–2026). "Yes" = a mandatory gate before "done". "Partial" = a gate exists but is not mandatory, or is explicitly not a test replacement.

TOOL	MANDATORY VERIFICATION GATE BEFORE "DONE"?
Factory	**Yes** — Missions end each milestone with a validation phase: build, test, lint, typecheck, browse the app.
Amp	**Yes** — Ship runs the full test suite before push.
Devin	**Partial** — video QA, self-described "sanity check… not a replacement" for tests/CI.
Copilot Coding Agent	**Partial** — tests/linters from a repo workflow; outer gate is normal PR CI.
v0 (Vercel)	**Partial** — the Vercel build/CI gate only.
Claude Code	**No** — hooks only; the model may skip them.
Cursor	**No** — "the user is still responsible for verifying".
Jules	**No** — whatever the user's setup script specifies.
Replit	**No** — App Testing is on-demand, limited to certain stacks.
Lovable	**No** — verification tools "mostly run only when asked".
Bolt	**No** — no documented always-on gate.
Windsurf (Cascade)	**No** — the only default check is lint auto-fix.

> **The structural insight.** The tools converge on *running the code*, but almost none enforce a gate *by default* — enforcement is a hook the model may skip. A required check matched **by name**, where a skipped or neutral result is a fail, is the entire difference. And none of the twelve document proving that their *checks themselves* can fail — the mutation test.

The comparison — where we are ahead, and where we are not

Our own measured assessment against the live system (October 2026), not a vendor's brochure. Where the field is ahead, it says so.

DIMENSION	US	THE FIELD
Verification / anti-false-confidence	**9.5/10 — ahead**	Only Factory and Amp gate verification; none prove their checks can fail — ours do, with 502 recorded mutation tests and the unproven backlog down 44 → 5.
Model & pipeline measurement	**9/10 — ahead**	A/B-tested routing; reviewer never grades its own work.
Architecture	**8/10 — convergent**	Same planner → coder → reviewer shape as Factory's Missions.
Cross-session memory	**8/10**	Operational depth (redaction, in-flight board) with no commercial equivalent.
Distributed compute mesh	**Rare**	Git-as-bus, multi-OS, trust-ladder — no funded equivalent.
Product / adoption	**2/10 — pre-scale**	Devin, Cursor, Replit — funded, adopted.
Robustness / ops / security	**4/10 — solo**	Funded teams, on-call, compliance.

The closest commercial peer is Factory — ~219 engineers, ~$420M raised (third-party). A solo operator matches their architecture and exceeds their verification rigour **on the one dimension that decides trust**. That is the precise, narrow claim — not "better than Factory everywhere", but **ahead on the dimension nobody else has productised**.

Why it is defensible

The abstract idea — "make 'done' mean 'verified'" — is free and will be copied the week the spec ships. What cannot be copied cheaply is the **content**:

- **130 automated guards** — proven by **502 recorded mutation tests**, each injecting a seeded fault and confirming the guard goes red. The unproven backlog has been ratcheted down from **44 to 5**.
- **A 26-entry breakage ledger** — every self-inflicted failure filed with a detection rule and fed back to the reviewer every turn.
- **Measured model routing** — planner / coder / reviewer on separate, A/B-tested models, and a reviewer that can never grade its own work.

Those are bought with *incidents*, not with reading — they are reproducible only by living through the failures. The go-to-market mirrors that:

> **Three layers.** The **specification** is open (Apache-2.0) so the standard spreads; the **reference SDK** is source-available and licensed; the **guard corpus + hazard library** is the commercial subscription. Open the format, keep the enforcement and the incidents.

The disruptor thesis — and its honest boundary

"The winner in this market is decided by **who can make 'done' mean 'verified'**."

The bull case. Generation got cheap; trust did not. The first product to make verification a hard, named gate — and to certify it — owns the trust layer of the whole category, the way a SOC 2 owns compliance credibility. We already ship the discipline: 130 guards with 502 recorded mutation proofs (unproven backlog ratcheted 44 → 5), a reviewer that never grades its own work, a 26-entry hazard ledger, and a certification program that is version-scoped and revocable.

The honest boundary. The discipline is real and running; the *business* is pre-revenue and solo. What is true: the verification edge is genuine and cited to primary sources. What is *not* yet true: product-market-fit, traction, or a moat that survives a funded competitor reading the spec. An investor is buying the discipline, the first-mover position, and the running system — **not** a proven revenue curve.

> **Stated plainly, because it is the one thing we will not inflate:** we are not claiming "better than Factory" across the board. We are claiming **ahead on the trust layer, today, measured against primary sources** — and that the trust layer is the market's unclaimed ground.

Next steps

The system, the receipts, and the full 12-tool survey are ready to share under NDA.

- **The page:** morpheus.nz/verification
- **The open spec + reference:** github.com/robmitchellengineering-hub/valiant-verification
- **The receipts:** the 12-tool survey (verification-gap.md, cited to vendor docs)
- **Contact:** robmitchellengineering@gmail.com

License. Integrate. Acquire. Seeking the first design partners — AI-tooling teams and dev agencies shipping agent-generated work — to integrate the layer and earn the first references.

Sources. Anthropic on the verification bottleneck — claude.com/blog/running-an-ai-native-engineering-org. Tool-by-tool verification behaviour — vendor documentation cited in the

> accompanying survey (research/ai-coding-agents-internals-2025-2026.md). Financial figures — Sacra and press, third-party, point-in-time. Guard/hazard counts — measured against the live codebase, October 2026, and self-updating on morpheus.nz/verification.
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